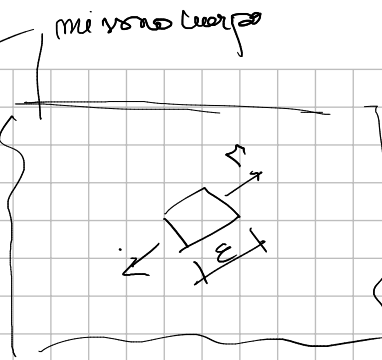
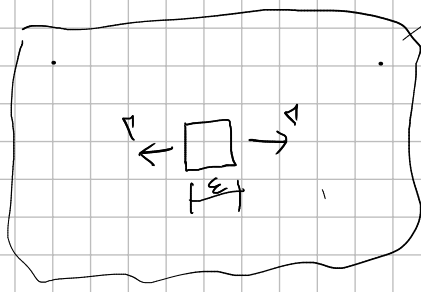
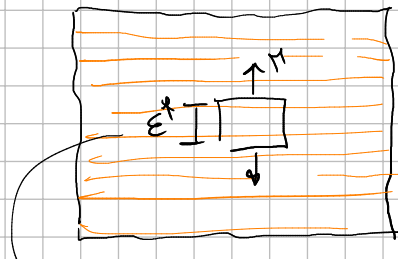
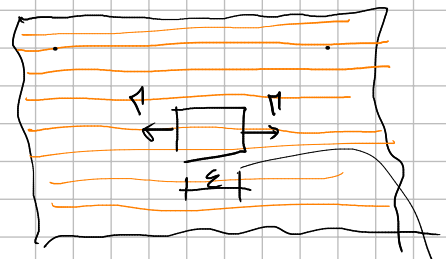




$$\underline{\underline{\tau}} = \underline{\underline{\sigma}} \cdot \underline{\underline{n}}$$

$$\underline{\underline{\tau}} = \underline{\underline{\sigma}} : \underline{\underline{\underline{\epsilon}}}$$

⇒ tensores constitutivos
(tensores 4to ordem)
sem isotrópicos



modelo no os
isotrópica
(isotrópica)

⇒ tensores constitutivos
no os isotrópicos

sem iguais

$$\lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + \nu (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}),$$

$$u_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}).$$

→ mudanças de τ_{ij} e

$$\underline{\underline{\tau}} = \underline{\underline{\sigma}} : \underline{\underline{\underline{\epsilon}}}$$

$$\sigma_{ij} = \lambda e_{\alpha\alpha} \delta_{ij} + 2\mu e_{ij}.$$

$$\underline{\underline{\sigma}} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$$

$$\underline{\underline{\tau}} = \underline{\underline{\sigma}} : \underline{\underline{\underline{\epsilon}}} \Rightarrow$$

$$\tau_{ij} = [\lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})] \epsilon_{kl}$$

$$= \lambda \delta_{ij} \cdot \epsilon_{kk} + \mu (\epsilon_{ij} + \epsilon_{ji}) =$$

$$2\mu \epsilon_{ij}$$

